HEARTS

Compendium of essential clinical terminology for hypertension-control programmes



World Health Organization

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Abbreviations

ABPM	ambulatory blood pressure monitoring
AOBP	automated office blood pressure
ACE inhibitor (ACEi)	angiotensin-converting enzyme inhibitor
ARB	angiotensin receptor blocker
BP	blood pressure
BPMD	blood pressure measuring device
CDS	clinical decision support
CVD	cardiovascular disease
DASH	dietary approaches to stop hypertension
DBP	diastolic blood pressure
EOD	end-organ damage
HBPM	home blood pressure monitoring
HEARTS	healthy-lifestyle counselling; evidence-based treatment protocols; access to essential medicines and technology; risk-based CVD management; team-based care; systems for monitoring
HTN	hypertension
LTFU	loss to follow-up
NCD	noncommunicable disease
SBP	systolic blood pressure
SPC	single pill combination
TLC	therapeutic lifestyle changes
WHO	World Health Organization

1. Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, and hypertension is the most important modifiable risk factor contributing to premature morbidity and mortality *(1)*. According to the WHO *Global report on hypertension 2025*, about 1.4 billion adults live with hypertension *(2)*. Effective detection, treatment, and long-term control of elevated blood pressure are essential components of population-level CVD prevention.

The WHO HEARTS technical package provides a comprehensive, evidence-based framework for improving CVD prevention and hypertension control in primary health care *(3)*. It does this by promoting eight interrelated technical components *(3)*:

- healthy-lifestyle counselling
- evidence-based treatment protocols
- access to essential medicines and technology
- risk-based CVD management
- team-based care
- systems for monitoring
- implementation guide
- tool for the development of a consensus protocol for treatment of hypertension



Successful implementation of the HEARTS framework requires not only standardized clinical protocols, but also harmonized terminology across service delivery, monitoring systems, quality improvement processes, and programme reporting. Inconsistent use or interpretation of key terms such as “patients under care”, “blood pressure control”, “loss to follow-up”, “cohort control”, or “standardized treatment protocol” can undermine indicator comparability, impede performance assessment, and weaken accountability mechanisms *(1, 3)*. This compendium responds to the need for a structured and standardized set of definitions to support consistent implementation of hypertension-control programmes within the HEARTS framework.

This compendium provides a consolidated reference for essential clinical, operational, and programme-related terminology used in hypertension-control programmes, aligned with the WHO HEARTS technical package and WHO guidelines on the pharmacological treatment of hypertension.

Specifically, this compendium aims to:

- support consistent interpretation and application of HEARTS-related terminology across countries and health-system levels;
- strengthen the “Systems for monitoring” component by clarifying operational definitions of core programme indicators;
- facilitate implementation of “Evidence-based treatment protocols” through standardized clinical terminology;
- enhance programme performance assessment, cohort monitoring, and quality-improvement processes;
- promote harmonization of reporting across national and subnational hypertension programmes.

The definitions are intended for adult hypertension-control programmes implemented as part of the HEARTS framework and may be adapted to national contexts while maintaining conceptual integrity.

This document is intended for:

- national and subnational programme managers implementing HEARTS;
- primary health-care clinicians and multidisciplinary teams;
- monitoring and evaluation officers;
- health-information system specialists;
- policy-makers and technical partners supporting CVD prevention.

By standardizing terminology, this compendium aims to strengthen implementation fidelity, improve indicator comparability, and support continuous quality improvement within HEARTS-based hypertension-control programmes.

2. How to use this compendium



This compendium is organized into three categories:

- **Clinical terminology**: terms used in diagnosis, risk assessment, treatment and patient follow-up. These support consistent clinical decision-making and documentation.
- **Operational terminology**: terms describing service-delivery processes, screening, registration, continuity of care, and quality improvement. These support programme implementation and management.
- **Monitoring and evaluation terminology:** indicator definitions used for reporting and performance assessment. Where applicable, these specify numerator, denominator and time reference to promote consistent measurement.

Users should first identify whether a term is clinical terminology, operational terminology or indicator-related to ensure appropriate application. Countries may adapt terminology to national contexts, but conceptual definitions should be preserved to maintain comparability across settings. Regular review is recommended to ensure continued alignment with updated WHO guidance and HEARTS implementation materials.

3. Methodology



The methodology used for this compendium aimed to ensure conceptual consistency, technical accuracy, and alignment with existing WHO recommendations and monitoring frameworks. The development process consisted of four main stages: (1) identifying relevant source documents; (2) selection and prioritization of terms; (3) drafting and harmonization of definitions; and (4) technical review and refinement.

Terminology was identified by reviewing key WHO normative and implementation documents related to CVD prevention and hypertension control, including but not limited to:

- WHO guidelines for the pharmacological treatment of hypertension in adults *(4)*;
- The WHO HEARTS technical package and associated implementation guides *(3)*;
- WHO cardiovascular risk-prediction charts and related materials;
- WHO noncommunicable disease (NCD) global monitoring frameworks and indicator metadata *(5)*;
- Relevant WHO publications on adherence *(6)*, essential medicines *(7)*, and primary health care *(8)*.

Where appropriate, definitions were aligned with established WHO guidance. In cases where no formal WHO definition was available, terminology commonly used in programme implementation was defined operationally to promote clarity and consistency.

Draft definitions were reviewed internally by technical staff with expertise in hypertension management, primary health-care implementation, and monitoring and evaluation. The review process focused on consistency with WHO guidance; internal coherence across sections; clarity of language; and applicability across different health-system contexts, particularly in low- and middle-income countries. Feedback from the review process was incorporated to refine definitions, resolve inconsistencies, and eliminate duplication.

This compendium is intended as a reference document to support standardization and does not replace WHO clinical guidelines or national regulatory frameworks.

4. Clinical terminology *(9–11)*

4.1 Diagnosis and classification

Hypertension (HTN) (high blood pressure):

Condition in which the pressure in the blood vessels is too high (140/90 mmHg or higher). This includes having a systolic blood pressure (SBP) ≥ 140 mmHg, or diastolic blood pressure (DBP) ≥ 90 mmHg, or taking blood pressure-lowering medication.

Elevated blood pressure

A SBP between 130–139 mmHg, or DBP between 80–89 mmHg.

Hypertensive emergency

Severe elevation in blood pressure, typically exceeding 180/120 mmHg, accompanied by signs of target-organ damage, such as injury to the brain, cardiovascular system, or kidneys.

White-coat hypertension

White-coat hypertension, also known as white coat syndrome, is a condition where a patient's blood pressure is elevated in a clinical setting but normal at home or during ambulatory blood pressure monitoring (ABPM). This phenomenon is often attributed to anxiety experienced during a medical visit, leading to higher readings in the presence of health-care professionals.

Masked hypertension

Masked hypertension, also known as white coat normotension, is defined as a condition where a patient's blood pressure readings appear normal in a clinical setting (e.g. doctor's office) but are elevated outside of that setting, such as at home or during daily activities. This phenomenon occurs when a person has normotensive office blood pressure (OBP) while experiencing hypertensive readings outside of it, often due to stress or other factors.

Resistant hypertension

Resistant hypertension is defined as blood pressure that remains above the therapeutic goal despite concurrent use of at least three antihypertensive agents of different classes, commonly including a long-acting calcium channel blocker, a blocker of the renin-angiotensin system (angiotensin-converting enzyme inhibitor or angiotensin receptor blocker), and a diuretic. Resistant hypertension is also diagnosed if blood pressure control requires four or more antihypertensive drugs.

4.2 Blood pressure measurement and monitoring

Systolic Blood Pressure (SBP)

SBP is the pressure in the blood vessels when the heart contracts or beats.

Diastolic Blood Pressure (DBP)

DBP is the pressure in the blood vessels when the heart relaxes between beats.

Blood pressure variability

This refers to the fluctuations in blood pressure readings over time, which can be influenced by factors such as heart rate, respiration, and sympathetic nervous system responses.

Ambulatory blood pressure monitoring (ABPM)

This is a non-invasive method used to measure blood pressure over a 24-hour period while the patient is in their own environment. This technique provides a true reflection of blood pressure throughout the day and night, making it a valuable tool for diagnosing and managing hypertension. ABPM involves taking blood pressure readings every 15–30 minutes during the day and every hour at night, allowing for a comprehensive assessment of blood pressure variability.

Home blood pressure monitoring (HBPM)

Consists of the measurement of blood pressure by a person outside of the clinical setting, at home, and is a validated approach for out-of-office blood pressure (BP) measurement. HBPM provides valuable data for diagnosing and managing hypertension.

Automated office blood pressure (AOBP)

This is a method of measuring blood pressure that utilizes automated devices to provide accurate and reliable readings, minimizing human error and the “white coat” effect.

Target blood pressure

This is the recommended blood pressure level to be achieved, which may vary depending on patient characteristics or comorbidities. WHO recommends a target blood pressure treatment goal of:

- < 140/90 mmHg in all patients with hypertension without comorbidities;
- <130 mmHg in patients with hypertension and known CVD;
- <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease) *(4)*.

4.3 Risk stratification and pathophysiology

Primary (essential) hypertension

Primary hypertension, also called essential or idiopathic hypertension, is high blood pressure without a specific identifiable cause, defined by WHO as a SBD consistently ≥ 140 mmHg and/or a diastolic blood pressure consistently ≥ 90 mmHg, measured on two separate days.

This form of hypertension is termed primary or essential because there is no underlying identifiable medical condition causing it, unlike secondary hypertension, which has specific causes such as kidney disease or hormonal disorders.

Secondary hypertension

Secondary hypertension, as defined by WHO, refers to arterial hypertension due to an identifiable cause. It affects approximately 5–10% of the general hypertensive population and can be caused by various conditions such as kidney disease, endocrine disorders, and obstructive sleep apnoea. Unlike essential hypertension, which has no identifiable cause, secondary hypertension often responds well when the underlying condition is addressed.

End-organ damage (EOD)

This is a structural and/or functional injury in organs commonly affected by hypertension and vascular disease (e.g., heart, brain, kidneys, retina, and large arteries), attributable to chronic elevated blood pressure and related risk factors.

Atherosclerosis

Atherosclerosis is a common disease that develops when a sticky plaque builds up in the arteries. Diseases linked to atherosclerosis are the leading cause of death worldwide.

Sympathetic overactivity

This is a sustained increase in activity of the sympathetic nervous system (e.g., increased sympathetic nerve firing and/or circulating catecholamines) that can contribute to higher blood pressure and cardiovascular strain.

Cardiovascular-risk score

The cardiovascular-risk score is a calculated estimate of an individual's absolute probability of experiencing a major cardiovascular event over a defined time horizon, typically expressed as a percentage risk over 10 years. WHO has developed 10-year CVD-risk charts to assess CVD risk among people aged 40 years and over (lab-based and non-lab-based).

4.4 Clinical assessment and management

Baseline evaluation

This is defined as the initial clinical assessment at the start of care, used to establish diagnosis, identify comorbidities and complications, and set a reference point for follow-up and treatment planning.

Risk-factor profiling

This refers to the systematic identification and documentation of cardiovascular-risk factors (e.g., smoking, diabetes, lipids, obesity) to guide total cardiovascular-risk reduction alongside blood pressure management.

Treatment adherence

This is the extent to which a person's behaviour (taking medication, following dietary advice, and/or making lifestyle changes) corresponds with agreed recommendations from a health-care provider *(6)*.

Effective antihypertensive treatment

Effective antihypertensive treatment refers to the combination of evidence-based pharmacological and non-pharmacological interventions (based on WHO or national recommendations, protocols or guidelines) that achieve and maintain blood pressure control to recommended targets, reduce cardiovascular risk, and prevent hypertension-related morbidity and mortality.

Lifestyle assessment

This is a structured assessment of behavioural and environmental factors that influence cardiovascular risk and blood pressure, e.g., diet, including salt intake, physical activity, tobacco, alcohol, weight etc.

Therapeutic inertia

This can be defined as a failure of clinicians to intensify antihypertensive therapy when blood pressure treatment goals are unmet (≥ 140/90 mmHg), even when the patient has been seen two or more occasions. This contributes significantly to uncontrolled hypertension and often occurs because of concerns about side effects such as hypotension; assumptions that a slightly lower reading represents improvement; patient reports of missed doses; or competing priorities during the clinical encounter.

Diagnostic inertia

Diagnostic inertia is the failure of health workers to diagnose hypertension in adults who consistently present with elevated blood pressure (≥ 140/90 mmHg), even when they have been seen twice (or more) in health facilities. It represents a critical lapse in care in which clinicians miss the opportunity to identify and initiate treatment. This failure contributes significantly to poor population-level blood pressure control and can have life-threatening consequences, as missing a diagnosis of hypertension can be as dangerous as missing other serious medical conditions.

Titration

Titration refers to stepwise dose adjustment and/or addition of medicines over time to achieve target blood pressure while monitoring tolerance and safety.

Side-effect profile

This is the expected pattern and frequency of adverse effects associated with a medicine or regimen, used to guide selection and patient counselling.

Polypharmacy

Polypharmacy is the use of multiple medicines by one person (often to manage multimorbidity), which can increase regimen complexity and the risk of adverse events and non-adherence if not regularly reviewed.

Contraindications

Contraindications are the specific clinical circumstances in which a medicine should not be used because the risk of harm outweighs potential benefit.

First-line antihypertensives

These are the classes of drug recommended for initial pharmacological treatment of hypertension. WHO's 2021 guideline for the pharmacological treatment of hypertension in adults recommends (as first-line antihypertensive drugs): thiazide/thiazide-like diuretics, ACE inhibitors or ARBs, and long-acting dihydropyridine calcium-channel blockers.

Combination therapy

Combination therapy refers to the use of two (or more) antihypertensive medicines (drugs) together to achieve blood pressure control, preferably initiated as a combination when pharmacological treatment is required.

Single-pill combination

This is a fixed-dose combination of two or more antihypertensive medicines in a single pill, used to simplify treatment and improve adherence and persistence.

4.5 Healthy-lifestyle and non-pharmacological interventions

Dietary approaches to stop hypertension (DASH)

DASH is a flexible, balanced eating plan designed to support a heart-healthy dietary pattern and help lower blood pressure, emphasizing fruits/vegetables/whole grains and less sodium.

Sodium restriction

This involves reducing dietary sodium intake to recommended levels to lower blood pressure and cardiovascular risk; for adults, WHO recommends < 2000 mg/day sodium (≈< 5 g/day salt).

Therapeutic lifestyle changes (TLC)

Refers to a structured lifestyle programme combining diet, physical activity, and weight management to improve cardiovascular risk.

Behavioural counselling

Behavioural counselling is a patient-centred approach that supports individuals to adopt and sustain healthier behaviours (e.g., diet, physical activity, tobacco cessation), typically using structured brief interventions and follow-up.

5. Programme operations terminology *(12–14)*



5.1 Service delivery and health systems capacity

Guideline

This broadly refers to comprehensive, evidence-based documents developed by expert bodies (WHO/others) that contain recommendations for clinical practice or public health policy. They describe recommended drug classes, multiple treatment options, and general principles for dose titration. These guidelines are designed to help end-users make informed decisions on specific actions, such as clinical interventions or public health measures, aiming to achieve the best possible health outcomes. They are often lengthy and best suited for specialist interpretation and clinical decision-making.

Standardized treatment protocol

Typically simple, one-page documents, these protocols are short, clear, and uniform sets of clinical instructions that specify the exact medicines, doses, and step-by-step titration needed to manage a condition for primary care, making them easy to use and suitable for national or subnational scale-up. They represent an adaptation of guideline evidence to a specific health system by providing short, clear, standardized instructions.

Availability of hypertension core medicines

This sets out the extent to which essential antihypertensive medicines (as defined by a programme's "core" list) are in stock and available at the point of care on the day of assessment/survey. Availability of hypertension medicines ensures an uninterrupted supply of essential medicines, which in turn supports patient adherence to care.

Validated blood pressure devices

A validated device is one that has undergone independent accuracy testing by an organization such as an accredited testing laboratory or academic institution, using publicly available, internationally accepted protocols (e.g., ISO 81060-2). This process involves rigorous clinical comparison of the device model against a reference standard, among real patients. Validation is performed once per device model (not for each individual unit), and demonstrates that the model can reliably measure blood pressure. Proof of validation is required before procurement.

Accurate blood pressure measuring devices

Accurate blood pressure measuring devices (BPMDs) are those that have successfully passed rigorous, independent validation testing against a recognized reference standard and are available with a range of appropriate cuff sizes to ensure correct measurement across different arm circumferences.

Calibrated blood pressure devices

Calibration is a routine maintenance procedure performed on each individual device to verify and adjust its readings against a known standard over time. For manual (aneroid) devices, this typically involves adjusting the pressure gauge so that the needle accurately reflects true cuff pressure when compared with a mercury column or calibrated digital reference. Calibration must be repeated periodically (e.g., every 6–12 months) to ensure continued measurement accuracy. Automated BPMDs do not require user calibration, as they use built-in sensors and algorithms. Instead, periodic inspection is recommended to ensure that the cuffs and tubes are intact, and sensors are functioning properly. Only authorized agencies should be contracted for repairs or servicing.

Recommended blood pressure devices

These are automated, upper-arm cuff blood pressure measuring devices that meet quality and accuracy requirements (including clinical validation) for use in screening, diagnosis and management.

Availability of a functional blood pressure measuring device

This means the presence of at least one recommended blood pressure measuring device that is operational and can be used to measure blood pressure at the time of assessment (including necessary cuffs/power/working condition).

Supportive supervisory visit

This is a planned, collaborative, on-site visit where supervisors work with facility teams to guide, mentor, and solve problems together to improve the quality of hypertension care. Instead of fault-finding, these visits focus on correct BP measurement, adherence to treatment protocols, medicine availability, accurate data, and effective patient follow-up. Using clear objectives and standardized tools, supportive supervision helps strengthen performance and improve blood pressure control.

Clinical decision support (CDS)

This includes tools (paper-based or digital) that provide person-specific, evidence-based recommendations to health workers at the point of care (e.g., treatment algorithms, prompts, risk tools) to support clinical decisions.

Care pathways

Care pathways represent a structured, protocol-based sequence-of-care processes that standardize diagnosis, treatment, follow-up, and referral for hypertension/CVD risk management to improve quality and outcomes. They are multidisciplinary tools designed to optimize care delivery by standardizing the sequence and timing of key interventions. They aim to improve service efficiency, enhance coordination among care teams, and promote consistent, high-quality care. Care pathways reduce unnecessary practice variation, address therapeutic inertia, support timely and appropriate clinical decisions, optimize resource use, and strengthen health-system performance.

5.2 Screening and case finding

Opportunistic screening

This refers to screening for hypertension as part of routine blood pressure measurement among all adults who visit health facilities for routine health service encounters, enabling the detection of elevated blood pressure/hypertension and a pathway towards diagnosis and care.

Case finding

This involves provider-initiated activities (e.g. mandatory, accurate BP measurement) to identify people with a condition (or at high risk of one) such as hypertension, who may not yet be diagnosed, followed by confirmatory diagnosis and linkage to care.

Assessment of CVD risk

This is an estimation of the 10-year absolute CVD risk among adults ≥ 40 years using a validated CVD risk-assessment tool such us the WHO CVD risk prediction charts/models to guide risk-based management.

5.3 Registration and continuity of care

Registrations

This is the number of patients with hypertension enrolled under the programme in a reporting period after diagnosis; includes those already on treatment and those newly diagnosed through screening.

Cumulative number of patients enrolled in care

This is the total number of patients registered/enrolled since programme initiation up to the reporting date (sum of registrations over time).

Patients under care

These are the patients with hypertension who had at least one visit (new or follow-up) to a health-care facility in the 12 months before the reporting date.

Missed visit

These are the registered patients with hypertension who missed their last appointment in the past 3 months.

Loss to follow-up (LTFU)

These are registered patients with hypertension who did not have a single follow-up visit in the 12 months before the reporting date.

6. Monitoring and evaluation definition and indicators



Indicator definitions are clearly presented in WHO's **Noncommunicable disease facility-based monitoring guidance *(15)***.

For each indicator, the document explicitly specifies the numerator and denominator, ensuring consistent interpretation and facilitating standardized measurement across facilities

Hypertension new patients

Number of people with hypertension who were newly registered during the past 3 months.

Hypertension treated patients

Number of people with diagnosed hypertension who were treated with antihypertensive medication during the reporting period.

Hypertension patient retention

Proportion of established patients with hypertension with at least one visit in the past 12 months.

Blood pressure under control

This is defined as SBP below 140 mmHg (< 140 mmHg) and DBP below 90 mmHg (< 90 mmHg) at 3–6 months after treatment initiation.

Blood pressure control among people with hypertension

This refers to achieving a defined blood pressure threshold for "controlled" blood pressure based on WHO or national treatment protocol/guidelines, among the total number of people registered for hypertension treatment in the facility, excluding those who were newly diagnosed within the past 3 months.

Based on WHO guidelines, BP is considered controlled when:

- SBP < 140 mmHg and DBP <90 mmHg
- SBP < 130 mmHg among people with history of CVD
- SBP < 130 mmHg among high-risk people with hypertension, i.e., those with high CVD risk, diabetes mellitus, chronic kidney disease

Facility cross-sectional blood pressure control

This indicator refers to the proportion of patients under care at a facility whose blood pressure was controlled at their most recent recorded visit within a defined period (cross-sectional snapshot). The indicator covers the total number of people registered for hypertension treatment in the facilities whose BP was controlled at the last clinical visit in the reporting period.

Blood pressure not under control

This indicator is defined as a systolic BP equal to or above 140 mmHg (≥ 140 mmHg) or DBP equal to or above 90 mmHg (≥ 90 mmHg) at 3–6 months after treatment initiation.

Quarterly blood pressure control

This indicator is the proportion of patients registered for hypertensive treatment during a quarter (cohort) whose blood pressure is controlled 3–6 months after registration (measured quarterly).

Community-level blood pressure control

This is the proportion of the estimated population with hypertension in a defined geographic area (e.g., district/state) with controlled blood pressure (measured annually).

Cohort blood pressure control

This is the proportion of a defined registration cohort whose blood pressure is controlled at a specified follow-up time (“window”) after registration.

Community cross-sectional blood pressure control

This is the proportion of the estimated population with hypertension in a geographic area whose blood pressure is controlled, as measured at a defined time point/period (cross-sectional).

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